

SVKM'S NMIMS
Mukesh Patel School of Technology Management & Engineering
Department of Computer Engineering
Java Lab

Subject- Object Oriented Programming through JAVA

EXPERIMENT NO. 1

Aim: Study of Basic Java Programming Constructs

Materials:

1. Editor – Notepad/Notepad++
2. Command Prompt

Theory:

Variable: A variable is a place where the program stores data temporarily. As the name implies the value stored in such a location can be changed while a program is executing (compare with constant).

Type	Description	Default	Size	Example	Range
boolean	Logical values	false	Not JVM-defined	true, false	true or false
byte	8-bit signed integer	0	1 byte	10	-128 to 127
char	16-bit Unicode character	\u0000	2 bytes	'A', '\u0041'	0 to 65,535
short	16-bit signed integer	0	2 bytes	2000	-32,768 to 32,767
int	32-bit signed integer	0	4 bytes	1000, -500	-2,147,483,648 to 2,147,483,647
long	64-bit signed integer	0L	8 bytes	123456789L	±9.22e18
float	32-bit floating point	0.0f	4 bytes	3.14f	~6–7 digits precision
double	64-bit floating point	0.0d	8 bytes	3.14159d	~15–16 digits precision

```
class Example2 {
```

```
public static void main(String args[]) {
    int var1; // this declares a variable
    int var2; // this declares another variable
    var1 = 1024; //this assigns value to var1
```

```
E:\N068\LAB1>java Example2.java
var1 contains 1024
var2 contains var1 / 2: 512
```

Constants and literals:

```
public static void main(String args[]) {
    int var; // this declares an int variable
    double x; // this declares a floating-point variable

    var =

}}
```

```
manyanayak@Manyas-MacBook-Air Lab1 % javac Example8.java
manyanayak@Manyas-MacBook-Air Lab1 % java Example8.java
Original value of var: 10
Original value of x: 10.0

var after division: 2
x after division: 2.5
```

foundation in Java programming.

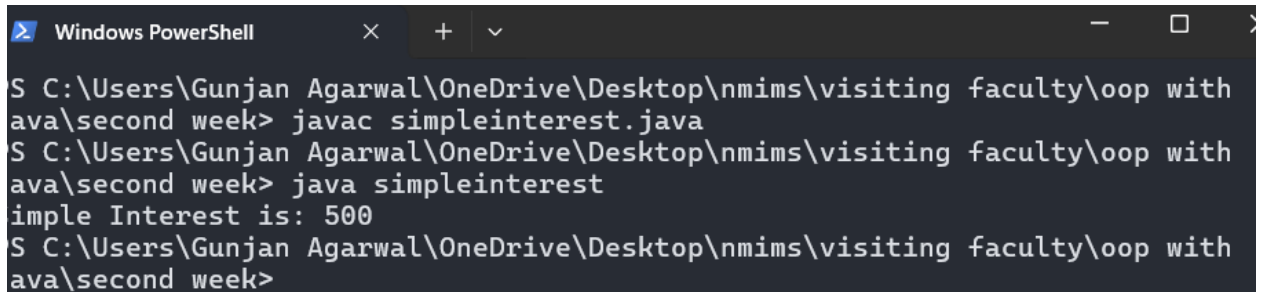
Post Lab: Any Five

1. Write a Java program to calculate simple interest using constants.
2. Write a Java program to convert temperature from Celsius to Fahrenheit.
3. Write a Java program to demonstrate the use of primitive data types.
4. Write a Java program to demonstrate relational operators.
5. Write a Java program to find the largest of two numbers using a ternary operator.
6. Write a Java program to check whether a number is even or odd using operators.
7. Write a Java program to calculate the sum of digits of a number.

Post Lab questions:

1. Simple interest

```
class Lab1_pl1{
    public static void main(String[] args){
        int p=1000;
        int r=5;
        int t=10;
        int SI;
        SI=(p*r*t)/100;
        System.out.println("The simple interest is:" + SI);
    }
}
```



```
Windows PowerShell
S C:\Users\Gunjan Agarwal\OneDrive\Desktop\nmims\visiting faculty\oop with
ava\second week> javac simpleinterest.java
S C:\Users\Gunjan Agarwal\OneDrive\Desktop\nmims\visiting faculty\oop with
ava\second week> java simpleinterest
imple Interest is: 500
S C:\Users\Gunjan Agarwal\OneDrive\Desktop\nmims\visiting faculty\oop with
ava\second week>
```

2. Celsius to fahrenheit

```

class Lab1_pl2{
    public static void main(String[] args){
        int c=25;
        int f;
        f = (c * 9 / 5) + 32;
        System.out.println("The temperature in Fahrenheit is:" + f);
    }
}

```

```

Windows PowerShell
PS C:\Users\Gunjan Agarwal\OneDrive\Desktop\nmims\visiting faculty\oop with java\second week> javac ctof.java
PS C:\Users\Gunjan Agarwal\OneDrive\Desktop\nmims\visiting faculty\oop with java\second week> java ctof
emperature in Fahrenheit is: 77
PS C:\Users\Gunjan Agarwal\OneDrive\Desktop\nmims\visiting faculty\oop with java\second week> |

```

3. Check whether no is odd or even

```

class Lab1_pl3{
    public static void main(String[] args){
        int n=79;
        System.out.println("The number taken is:" + n);
        if(n%2==0){
            System.out.println("The number " + n + " is even.");
        }
        else{
            System.out.println("The number " + n + " is odd.");
        }
    }
}

```

```

Windows PowerShell
PS C:\Users\Gunjan Agarwal\OneDrive\Desktop\nmims\visiting faculty\oop wi java\second week> javac evenorodd.java
PS C:\Users\Gunjan Agarwal\OneDrive\Desktop\nmims\visiting faculty\oop wi java\second week> java evenorodd
Number is Odd
PS C:\Users\Gunjan Agarwal\OneDrive\Desktop\nmims\visiting faculty\oop wi java\second week>

```

4. Calculate the sum of digits of a number

```

class Lab1_pl4{
    public static void main(String[] args){
        int sum=0;
        int n=2360;
        int digit;
        for(;n>0;n=n/10){
            digit=n%10;
            sum = sum + digit;
        }
        System.out.println("The sum of digits of 2360 is " + sum);
    }
}

```

```

PS C:\Users\Gunjan Agarwal\OneDrive\Desktop\nmims\visiting faculty\
java\second week> javac sumofdigits.java
PS C:\Users\Gunjan Agarwal\OneDrive\Desktop\nmims\visiting faculty\
java\second week> java sumofdigits
The sum of digits of 2360 is 11
PS C:\Users\Gunjan Agarwal\OneDrive\Desktop\nmims\visiting faculty\
java\second week>

```

5. Find the largest no using a ternary operator

```

class Lab1_pl5{
    public static void main(String[] args){
        int a=68;
        int b=10;
        System.out.println("a=" + a + " and b=" + b);
        if(a>b){
            System.out.println("a is greater than b.");
        }
        else{
            System.out.println("b is greater than a.");
        }
    }
}

```

```

Windows PowerShell
PS C:\Users\Gunjan Agarwal\OneDrive\Desktop\nmims\visiting faculty\oop
java\second week> javac largestno.java
PS C:\Users\Gunjan Agarwal\OneDrive\Desktop\nmims\visiting faculty\oop
java\second week> java largestno
a = 68 and b = 10
a is greater than b.
PS C:\Users\Gunjan Agarwal\OneDrive\Desktop\nmims\visiting faculty\oop
java\second week>

```